Methanol Handling Procedure

Summary

Contents	Details
Regulatory Framework	Regulations
Methanol Properties and Hazards	Chemical properties
Indoor Handling	Ventilation, Personal Protective Equipment, Indoor-Specific Precautions
Outdoor Handling	Ventilation and Weather considerations, Personal Protective Equipment, Outdoor-Specific Precautions
Transportation of Methanol	Classification and Documentation, Placarding and Labelling, Vehicle and Driver Requirements, Loading and Unloading
Emergency Procedures	In Case of Spill, In Case of Exposure or Poisoning, Key Emergency Contacts

1. Regulatory Framework

This procedure has been developed in compliance with the following federal and Quebec provincial regulations and standards. All personnel handling methanol must be aware of these obligations.

Regulation/Standard	Applicability
SIMDUT 2015 (Hazardous Products Act & Hazardous Products Regulations)	SDS availability, labelling, worker training on hazardous products including methanol
RSST — Règlement sur la santé et la sécurité du travail (c. S-2.1, r. 13)	Quebec OHS standards: exposure limits, PPE, ventilation, first aid requirements
LSST — Loi sur la santé et la sécurité du travail (c. S-2.1)	Employer and worker obligations in Quebec workplaces
TDG Act — Transportation of Dangerous Goods Act, 1992	Classification, documentation, placarding, and emergency response for methanol transport (UN 1230)
TDG Regulations (SOR/2001-286)	Specific requirements for shipping documents, safety marks, and ERG procedures

Environmental Quality Act (c. Q-2) & ECCC Spill Reporting	Environmental spill reporting obligations in Quebec
NFPA 30 — Flammable and Combustible Liquids Code	Storage and handling of flammable liquids (reference standard)

- Under SIMDUT 2015 (aligned with GHS), methanol is a Class 3 Flammable Liquid and a Category 1 Acute Toxicity substance. A compliant Safety Data Sheet (SDS) must be accessible at all times in the work area.
- Under the LSST (art. 51), employers must take all necessary measures to protect the health, safety, and physical integrity of workers. This procedure constitutes a written safe work practice as required.
- Under the RSST (art. 40-44), the employer must ensure that the workplace meets ventilation standards and that workers are equipped with appropriate PPE when exposed to hazardous substances.
- CNESST (Commission des normes, de l'équité, de la santé et de la sécurité du travail) is the enforcement authority in Quebec. Non-compliance may result in stop-work orders and fines.

2. Methanol Properties and Hazards

Methanol (CH_3OH) is a colorless, flammable, and highly toxic liquid. Per SIMDUT 2015, it is classified as a Flammable Liquid (Category 3) and Acute Toxicity (Category 3 — oral, dermal, inhalation). The following properties are critical for safe handling:

- Flash point: 11°C (closed cup) — fire risk is present at room temperature
- Explosive limits: 6% to 36.5% in air (LEL/UEL)
- Vapor density: 1.11 (heavier than air — vapors accumulate at ground level and in low-lying areas)
- Toxicity: absorbed by inhalation, skin contact, and ingestion; as little as 10 mL can cause permanent blindness and 30 mL can be fatal
- RSST exposure limit (VEMP): 200 ppm (8-hour TWA) | Short-term exposure limit (VECD): 250 ppm (15-min)
- UN Number: 1230 | Packing Group: II | TDG Hazard Class: 3 (Flammable Liquid), subsidiary Class 6.1 (Toxic)
- SIMDUT-compliant SDS must be posted or readily accessible in the work area at all times (RSST art. 62.1) Performing the visual inspection

3. Indoor Handling

3.1 Ventilation

Ventilation is the primary engineering control for indoor methanol handling, as required under RSST art. 42. Since vapors are heavier than air, they accumulate at floor level and must be actively removed.

- Mechanical general ventilation is mandatory: minimum 10 air changes per hour in the work area
- Local exhaust ventilation (LEV) — fume hood or articulated extraction arm — must be used directly at the source during transfer or when opening containers
- Exhaust inlets must be positioned low (within 30 cm of the floor) to capture dense vapors
- Exhaust air must be discharged to the outdoors and must not be recirculated without filtration treatment
- Monitor vapor concentrations using a calibrated combustible gas detector (LEL sensor) during extended operations; the RSST VEMP is 200 ppm
- If ventilation fails or vapors exceed the RSST VEMP, stop work immediately, evacuate, and notify supervision

3.2 Personal Protective Equipment (PPE)

PPE selection must follow a hazard assessment as required under RSST art. 40. PPE is the last line of defense and must complement, not replace, engineering controls.

- Respiratory protection: half-face respirator with organic vapor (OV/AG) cartridges when ventilation is insufficient or during spills; SCBA required in emergency situations (RSST art. 45)
- Hand protection: chemical-resistant gloves — nitrile (≥0.38 mm), butyl, or neoprene; avoid latex and PVC (RSST art. 40)
- Eye and face protection: chemical splash goggles; face shield required if splashing is possible (RSST art. 41)
- Body protection: chemical-resistant apron; full protective suit for large-volume handling
- Antistatic safety footwear in compliance with CSA Z195 standard

3.3 Indoor-Specific Precautions

- Eliminate all ignition sources: switches, non-explosion-proof motors, open flames, cigarettes, and uncertified mobile phones
- All electrical equipment in the handling area must be rated for explosive atmospheres per CSA C22.1 (Canadian Electrical Code) Class I, Division 1 or 2
- Use only non-sparking tools (copper-beryllium or aluminum-bronze) in the handling area
- Bonding and grounding are mandatory between all containers during transfer to prevent static discharge (NFPA 30)
- Keep indoor methanol to the minimum quantity required; store excess in a ventilated, fire-rated cabinet per NFPA 30 and RSST art. 60
- Emergency eyewash station must be within 10 seconds of travel; emergency safety shower within 100 meters (RSST art. 195)

- CO_2 or dry chemical (ABC) fire extinguisher must be immediately accessible; never use water on methanol fires
- Post no smoking, eating, or drinking signs in the handling area (RSST art. 40)
- Work in pairs when handling more than 5 liters — never work alone in a closed indoor space with methanol
- All workers must complete SIMDUT 2015 training before handling methanol (Hazardous Products Act, s. 14)

4. Outdoor Handling

4.1 Ventilation and Weather Considerations

Outdoor environments generally provide better natural vapor dispersion, but certain meteorological conditions can create localized hazardous accumulations. A site assessment is required before every outdoor methanol operation.

- Check wind direction and speed before starting — workers must always position themselves upwind of the work area
- Avoid operations in calm conditions (wind below 1 km/h): vapors may accumulate without dispersing
- In hot weather (above 30°C), evaporation rate increases significantly — enhance respiratory protection and shorten exposure duration
- Be aware of adjacent low-lying areas (ditches, manholes, building vents) where dense vapors may accumulate
- Do not perform transfer operations in high winds (above 40 km/h): risk of uncontrolled splashing and vapor drift

4.2 Personal Protective Equipment (PPE)

- Respiratory protection: half-face OV cartridge respirator for small transfers with adequate wind; SCBA for large-volume operations or spill response
- Hand protection: nitrile or butyl chemical-resistant gloves (same requirements as indoors)
- Eye and face protection: chemical splash goggles and face shield; wind increases splash risk
- Clothing: antistatic cotton or treated workwear; avoid synthetic fabrics that generate static electricity

4.3 Outdoor-Specific Precautions

- Establish a safety perimeter of at least 5 meters around the handling area using barrier tape and warning signs
- Prohibit all ignition sources within a 15-meter radius: vehicles, engine equipment, unclassified generators, and smoking
- Ground and bond all containers and transport vehicles before any transfer — grounding cable is mandatory

- Spill containment is required: use an impermeable tarp or portable secondary containment berm under all transfer points — minimum capacity 110% of volume handled, as required under the Environment Quality Act (Quebec)
- Never work near open water, ditches, or storm drains without spill protection; methanol is fully miscible with water and is reportable under the Environment Quality Act (art. 21) if released to the environment
- Have a minimum 6 kg ABC or CO_2 fire extinguisher within 15 meters
- Use a shade canopy over open containers on hot days to reduce evaporation
- For nighttime operations, use explosion-proof lighting certified per CSA C22.1
- Carry a means of communication (radio or phone) and notify supervision before beginning operations

5. Transportation of Methanol (TDG)

The transportation of methanol by road, rail, marine, or air within Canada is governed by the Transportation of Dangerous Goods Act, 1992 and the TDG Regulations (SOR/2001-286). The following requirements apply to all personnel involved in shipping, receiving, or transporting methanol.

5.1 Classification and Documentation

- Proper Shipping Name: METHANOL | UN Number: 1230 | Hazard Class: 3 | Subsidiary Risk: 6.1 | Packing Group: II
- A completed shipping document (bill of lading or dangerous goods declaration) must accompany every shipment and be accessible to the driver at all times (TDG Reg. Part 3)
- The shipping document must include: UN number, proper shipping name, hazard class, packing group, total quantity, and emergency contact number (CANUTEC: 1-888-226-8832)
- 24/7 emergency response assistance number must be listed on the shipping document — CANUTEC (1-888-226-8832) is recommended

5.2 Placarding and Labelling

- Containers must bear SIMDUT 2015-compliant labels with GHS pictograms: flame (flammable liquid) and skull & crossbones (acute toxicity)
- Transport vehicles carrying 3,000 L or more of methanol must display TDG placards: Class 3 (Flammable Liquid) and Class 6.1 (Toxic) on all four sides of the vehicle (TDG Reg. Part 4)
- Containers must be UN-certified for the packing group (PG II) and must not be overfilled (max 95% capacity for liquids)

5.3 Vehicle and Driver Requirements

- Drivers must hold a valid TDG training certificate and be familiar with the Emergency Response Guidebook (ERG Guide 131 for methanol)

- Vehicle must carry: shipping documents, ERG (current edition), fire extinguisher, and appropriate spill containment equipment
- Methanol must not be transported with incompatible materials (oxidizers, strong acids, or alkali metals)
- Vehicles must be inspected before departure for container integrity, proper securing, and absence of leaks
- During transport, the vehicle must be parked away from populated areas and heat sources; engine must be off during loading/unloading

5.4 Loading and Unloading

- Bonding and grounding required during all loading and unloading operations
- Engine must be shut off during transfer; no smoking within 15 meters
- Loading/unloading must not take place near open flames, drains, or waterways without spill protection
- In Quebec, any spill during transport must be reported immediately to: Urgences-Environnement: 1-800-463-4311 and Transport Canada if on a federal road

6. Emergency Procedures

6.1 In Case of Spill

- Immediately evacuate non-essential personnel and secure the area
- Eliminate all ignition sources in the surrounding area
- Contain the spill using dry sand, vermiculite, or universal chemical absorbent — never use sawdust (combustible)
- Collect contaminated absorbent material in approved hazardous waste containers labeled per SIMDUT 2015
- Report any release to the environment immediately: Urgences-Environnement Québec: 1-800-463-4311 (as required under the Environment Quality Act, art. 21)
- Complete an internal incident report and notify CNESST if a worker was exposed or injured (LSST art. 62)

6.2 In Case of Exposure or Poisoning

- Inhalation: move the person immediately to fresh air; if breathing has stopped, begin first aid and call 911
- Skin contact: remove contaminated clothing immediately and wash with soap and water for at least 15 minutes (RSST art. 195)
- Eye contact: flush immediately with running water for at least 15 minutes while holding eyelids open; seek medical attention
- Ingestion: DO NOT induce vomiting — call Centre antipoison du Québec: 1-800-463-5060 or 911 immediately

6.3 Key Emergency Contacts

Organization	Contact
Emergency / Ambulance / Fire	911
Centre antipoison du Québec	1-800-463-5060
Urgences-Environnement Québec	1-800-463-4311
CANUTEC (TDG Transport Emergencies)	1-888-226-8832
CNESST (Workplace Incidents)	1-844-838-0808

This procedure must be reviewed annually or following any incident involving methanol. All personnel must complete SIMDUT 2015 training before handling methanol. This document must be kept accessible in the work area at all times as required under the RSST.

Prepared in compliance with: SIMDUT 2015 • LSST (c. S-2.1) • RSST (c. S-2.1, r. 13) • TDG Act, 1992 • Environment Quality Act (c. Q-2) • NFPA 30 • CSA C22.1